Mega XL — custom upgrades & build notes

Upgrades to a Kickstarter v1.0 Mega XL CNC router.

This documents the changes made to a first-generation (Kickstarter v1.0) Mega XL: a taller, stiffer Z, closed-loop motors, all-proximity homing, and a complete electronics rebuild around a grblHAL controller in a custom enclosure with a proper 48 V power system, air assist, and probing.

At a glance

Subsystem	Stock (v1.0)	Upgraded
Vertical clearance	stock	+3″ via Mark Smith riser kit
Z assembly	stock Z	Mark Smith Z gantry (only V-wheels + X motor mount reused)
Stepper motors	generic open-loop steppers	all-in-one closed-loop steppers (driver integrated in motor)
Limit / homing	mechanical limit switches	proximity sensors on every axis
Probing	—	touch plate + 3D probe in parallel; grounded spindle (no tool clip)
Controller	8-bit Arduino board	grblHAL Teensy 4.x (USB + Ethernet)
Electronics enclosure	cramped stock steel box	custom 12×16×6″ aluminum / plexiglass
Power	stock	48 V Meanwell system, E-stop, buck-derived 12 V / 5 V rails
Air assist	none	Mist-relay 12 V solenoid → 30 PSI nozzle at the cut

Frame & mechanical

- **Mark Smith riser kit** — adds **3″ of additional vertical (Z) clearance**, allowing taller stock and longer tooling.
- **Mark Smith Z gantry** — a **complete replacement of the Z assembly**. The only stock parts carried over are the **four V-wheels** and the **X stepper motor mount**; everything else is new.

Motors

The generic steppers that shipped with the Mega XL were replaced with **all-in-one closed-loop stepper motors** — each motor has its stepper driver built into the motor body. This removes the separate driver boards, adds closed-loop position feedback (lost-step detection / alarm), and simplifies wiring to a single cable per motor.

- **Individual enable** per motor — each stepper's `EN` line is driven separately.
- The motors' `AL` **(alarm) outputs are wire-OR'd together** and fed to the controller's `ST0` **input**, so a stall / lost-step fault on any motor signals the controller.

All four axes are closed-loop. The machine is a **dual-Y gantry** — the two Y gantries are driven by ganged motors (**Y + A**), one per side.

Axis	Holding torque	Drives
Y + A	3.2 N·m each	the two Y gantries (ganged)
X	2.0 N·m	spindle carriage along the cross-beam (rack & pinion)
Z	1.2 N·m	Z

All motors have an **8 mm output shaft with a flat**, with pinions/couplings retained by a grub screw on the flat. *Build note:* the stock Mega XL pinions were 6 mm-bore and were **reamed to 8 mm** at a machine shop — this thinned the set-screw boss and reduced grub-screw thread engagement. See the rack-and-pinion maintenance appendix.

Homing & limit sensing

The Mark Smith Z gantry uses a **proximity sensor** as its limit switch. To keep one sensing scheme across the machine, **all remaining stock mechanical limit switches were replaced with the same style of proximity sensor**, so every axis homes on identical, non-contact hardware.

Probing

- The **touch-plate input is wired in parallel with the primary probe input**, alongside the **3D mechanical probe** — either device can trigger the same probe input.
- The **spindle mount is grounded**, so the tool itself is at ground through the spindle. That completes the touch-plate circuit on contact — **no alligator clip on the tool** is required.

Controller & electronics enclosure

- Discarded the **cramped stock steel electronics enclosure** and the **stock 8-bit Arduino controller board**.
- Control is now a **grblHAL Teensy 4.x** controller on a breakout board (32-bit).
- Built a **custom 12″ × 16″ × 6″ enclosure**:
    - Frame: **2020 aluminum extrusion**.
    - Five sides: **0.0625″ (1/16″) aluminum sheet**.
    - Top: **1/8″ clear plexiglass** (visibility into the electronics).

Connectivity

The Teensy 4.x supports Ethernet, and the breakout board has the **Ethernet option installed**. Both links to the Teensy are brought out as **panel patch-through connectors on the side of the enclosure**: a **female USB-A** jack and a **female Ethernet (RJ45)** jack.

Power distribution

A single **220 V, 20 A branch circuit** feeds the machine through a **front-panel emergency-stop button** (the E-stop breaks the 220 V input, so it drops both the PSU and the spindle drive). The 220 V powers a **48 V Meanwell power supply** and the **VFD** (spindle drive). Everything else is derived from the 48 V rail through two buck converters.

220 V, 20 A → front-panel E-stop →

└─ 48 V Meanwell power supply

└─ VFD (spindle drive)

48 V (PSU output) →

└─ 4 × stepper motors

└─ 12 V / 3 A buck converter

└─ 5 V / 3 A buck converter

12 V (buck) →

└─ 4 × proximity limit sensors

└─ 12 V fan cooling the spindle coolant loop (runs while the spindle runs)

└─ grblHAL Teensy 4.x breakout board, 12 V input — the board uses 12 V to generate the **0–10 V PWM spindle-speed signal** to the VFD

5 V (buck) →

└─ breakout board 5 V input — board logic power, **separate from the Teensy** (the Teensy is powered over USB)

└─ 3D probe

Relay outputs & air assist

The breakout board provides **5 relay outputs**, each **jumper-selectable for 5 V or 12 V**. Only the **Mist** output is in use, driving the air-assist nozzle:

> Mist relay → 12 V solenoid valve (compressed-air inlet) → regulator @ 30 PSI → needle valve at top of Z gantry → 1/8″ copper airline → swivel nozzle aimed where the tool meets the stock

Connectors & terminations

Ferrules are crimped on all terminal-block connections — every stranded conductor landing on a terminal block is ferruled for a clean, captured, reliable termination.

Enclosure pass-throughs use **Molex Mini-Fit Jr panel-mount connectors**. The choice was deliberate: with roughly 60 terminations to make, the builder had far more confidence producing **~60 quality crimps** than ~60 quality solder joints on aircraft-style connectors.

Cable & wiring schedule

All runs use shielded cable.

Cable	Run	Conductors
18/6 shielded	enclosure → each stepper motor	6 (assignment depends on motor type — see below)
22/3 shielded	enclosure → Y-axis proximity switches	`12V`, `Sig`, `GND`
22/6 shielded	enclosure → X & Z proximity sensors + 3D probe (shared)	`12V`, `GND`, `XLim`, `ZLim`, `P`, `5V`

18/6 stepper cable — conductor assignment

Conductor	Closed-loop stepper	Discrete (open-loop) stepper
1	`48V`	`A+`
2	`GND`	`A-`
3	`DIR`	`B+`
4	`PUL`	`B-`
5	`EN`	`EN`
6	`AL`	`AL`

All four axes on this build are closed-loop, so they all use the left-column assignment (`48V` / `GND` / `DIR` / `PUL` / `EN` / `AL`). `DIR` / `PUL` are direction/step to the integrated drives; `EN` = enable, `AL` = alarm. The discrete `A±` / `B±` column is kept for reference only — the same 18/6 cable would serve an open-loop motor if one were ever fitted.

Rack-and-pinion drive — setup & maintenance

X (and the Y/A gantry drives) run on rack and pinion. After ~10 hours the X pinion crept **left** under load. A closed-loop motor can't see slip *downstream* of its shaft encoder, so a pinion loosening on the shaft is invisible until a fixed reference (the toolsetter at G59.3) exposes it. With the bores reamed 6→8 mm the set-screw boss is thin and grub-screw grip is the weak point; X slips first because it takes the most direct cutting side-load, but the other reamed pinions carry the same risk.

Pinion-to-shaft — stop the slip

1. Pull the pinion; confirm the grub screw seats on the shaft **flat** (not a round section), and check whether the screw marred the flat or only kissed it.
2. **Degrease** shaft, bore, and screw threads — oil defeats both grip and threadlocker.
3. Use the **longest grub screw** the thinned boss will take, on the flat; add a second screw if there's room.
4. **Threadlocker:** blue (242/243) is normally enough; with the reduced thread engagement here, **red (271)** is justified as insurance against back-out — just note removal then needs heat (~250 °C, easy on a bare pinion).
5. Torque to screw size (M3 ~1.0–1.4, M4 ~2.5–3, M5 ~5–6 N·m); snug, don't strip the hex. Let red cure ~24 h before heavy cuts.

Pinion-to-rack mesh & backlash

1. Loosen the motor mount so the pinion can move toward / away from the rack.
2. Shim **0.05–0.10 mm** (feeler gauge, or ~0.1 mm folded paper) between a pinion tooth and the rack, push the pinion in against the shim, tighten the mount, remove the shim.
3. Check the mesh over the **full travel** — it should stay lightly meshed everywhere. Tight at one end + slack at the other means the rack isn't parallel to the axis; realign the rack first.
4. Confirm full tooth-width engagement (pinion not riding a tooth edge).

Verify

1. By hand: smooth full travel, minimal lash; rock the gantry and feel for play at the pinion.
2. Dial indicator: commanded vs actual on small moves; reverse direction to read backlash (a few thou).
3. Re-run a heavy back-and-forth / cut loop with paint-pen **witness marks** (motor shaft ↔ pinion, and a pinion tooth ↔ a rack tooth) — no separation = fixed.
4. Confirm the `AL → ST0` alarm actually halts grblHAL (force a stall) so a real fault can't pass silently.

If it still creeps

A single grub screw — especially in a reamed, thin-wall boss — is marginal for rack-and-pinion loads. Durable fixes, in order: **proper 8 mm-bore pinions** (full set-screw boss) → **clamp / split-hub pinion** (grips the whole shaft, no set screw) → **keyed shaft**. Given the reamed pinions, the **clamp-hub style is the recommended upgrade** — it ends set-screw slip for good.